

Rohan Jain

jainrohanj@gmail.com | <https://www.linkedin.com/in/rohan-jain11/> | <https://github.com/rohanjain11>
<https://rohanjain11.github.io/Portfolio/> | +1 720-819-0419 | Vista, CA

EDUCATION

University Of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 - May 2026 | Boulder, USA

Mumbai University, Rajiv Gandhi Institute of Technology

Bachelor's of Engineering in Computer Engineering (GPA: 3.7/4)

Jul 2020 - May 2024 | Mumbai, India

SKILLS

Languages & Data: Python, R, SQL (MySQL, PostgreSQL; CTEs, window functions), MongoDB. **Analytics:** EDA, data cleaning, KPI definition, hypothesis testing, forecasting, clustering, association rules; evaluation (MAE, MSE, CRPS), baseline comparisons, error analysis. **Visualization & BI:** Tableau, Power BI, Excel (PivotTables, XLOOKUP, Power Query), Matplotlib, Plotly, Seaborn; dashboard design and stakeholder-ready reporting. **Data Quality & Workflow:** pandas, NumPy; validation gates (schema, unit, range), reproducible outputs, metric definitions and documentation. **Additional:** scikit-learn, FastAPI, REST APIs, Git/GitHub, Docker (basics), TensorFlow, PyTorch, Keras, LangChain, FAISS, Apache Spark; AWS (Lambda, S3), Azure.

EXPERIENCE

Data Science & Engineering Intern, Nexus Weather & Climate

May 2025 - Present | Boulder, CO

- Improved forecast KPI performance by replacing an older SVM bias-correction step with a tuned Random Forest, then validating results and writing rollout notes for review; reduced temperature MAE from 0.73 to 0.39 (47%) and humidity MAE from 4.68 to 2.37 (49%) using baseline comparisons.
- Addressed the need to communicate uncertainty by building a probabilistic CNN-LSTM with a custom CRPS loss, then producing clear plots and short decision notes for stakeholders; achieved a best reported CRPS of 0.1787 and supported risk-aware interpretation using simple, consistent reporting outputs.
- Standardized model evaluation by creating a repeatable training and benchmarking workflow with schema, unit, and range checks, then running SVR, RF, tuned RF, GBRT, XGBoost, and MLP under consistent rules; produced reusable analysis artifacts that track MAE and MSE against pre-ML baselines for transparent comparisons.
- Reduced rework by implementing an inference quality gate that accepts a model only when post-ML MAE is lower than the pre-ML baseline, preventing regressions from moving forward; reduced failed runs by 90% by automatically rejecting updates that did not meet baseline improvement criteria.
- Improved analysis of training stability by running controlled loss comparisons where only the loss changed (CRPS vs hybrid CRPS+MSE with $\alpha = 0.80$ vs Huberized CRPS) and logging runtime plus delta CRPS; hybrid ran 3.65 min vs 4.4 min baseline with mean delta -0.01 to -0.03, while Huberized showed -0.005 to -0.02 with steadier validation behavior.

Graduate Research Assistant, Kopf Lab, University of Colorado Boulder

Dec 2025 - Present | Boulder, CO

- Improved stable isotope data quality by standardizing parsing outputs across vendor formats, enforcing consistent field names, units, and schemas so the same file yields the same structure across platforms and package versions for repeatable analysis, reporting, and downstream comparisons.
- Reduced silent data integrity issues by adding fail-fast validation for corruption and edge cases using type, bounds, and structural checks (missing blocks, wrong ordering, truncation), returning clear errors instead of partial or incorrect outputs during ingestion and analysis workflows.
- Decoded legacy Thermo Isodat scan (.scn) files stored via MFC CArchive Serialize by mapping class layouts and field order, then implementing R readers that follow the same read sequence and reconstruct analysis-ready objects with traceable provenance for debugging, review, and consistent extraction.
- Strengthened auditability by attaching file IDs, format signals, parsing paths, and assumptions to outputs, and building regression tests with golden outputs plus corruption fixtures so updates do not break prior behavior or hide parsing errors during maintenance and iteration.
- Supported secure multi-user access planning by separating authentication from app execution, using Keycloak OIDC for login and role/group access control while ShinyProxy runs one container per user session, enabling safer operations, controlled access, and maintainable updates for lab users.

PROJECTS

SaferRide (Risk-Aware Bicycle Routing) | Python, FastAPI, SQL, PostgreSQL, PostGIS,

Sep 2025 - Dec 2025

GeoPandas, OSRM, Map UI

- Created SaferRide, a bicycle routing app that requests OSRM routes, offers multiple alternatives, and assigns a route-level risk score using crash density and hazard proximity; built a pipeline to clean city datasets, computed risk metrics with PostGIS spatial joins and GeoPandas, and served routes plus risk outputs via FastAPI with map overlays to compare safety versus time tradeoffs.

Denver International Airport Analytics | Python, Power BI, Tableau

Oct 2024

- Integrated ServiceNow and Azure DevOps data into analysis-ready pipelines, defined KPIs and built Power BI and Tableau dashboards with drill-downs for SLA status, queue times, and bottlenecks; authored reusable ETL queries and governance notes and KPI definitions for consistent refresh logic, reduced manual reporting by 40%, and earned 3rd place in a hackathon for dashboard impact.

AI PDF Chatbot | Python, FastAPI, React, LangChain, FAISS

Jan 2025

- Built a PDF question-answering tool for scanned and digital documents by combining OCR, embeddings, and FAISS retrieval behind a FastAPI and React UI; standardized chunking and prompt templates, supported secure uploads and real-time responses, added usage logging for query analytics, and ran labeled spot checks to refine retrieval quality so recurring research questions returned more relevant passages.

PUBLICATION

- Technical paper: *Yoga Posture Detection and Correction*. [Link](#)